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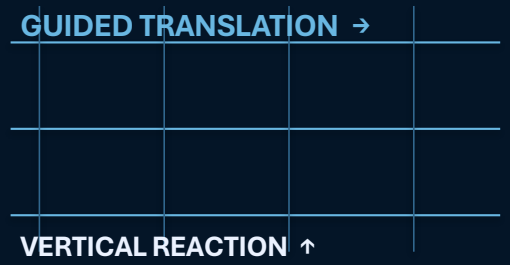
ENGINEERING PORTFOLIO / 2025 – 26

SLIDER SUPPORT

Pin-in-slot / Solid Mechanics

First-Year Mechanical Engineering Project | LSBU







BOKHTEAR UDDIN BADHON

BEng (Hons) Mechanical Engineering • London South Bank University

SEEKING 2027 INDUSTRIAL PLACEMENT

INNOVATIVE ENGINEERING.
A SUSTAINABLE TOMORROW.

Turning statics into something tangible

A first-year team project investigating the mechanical behaviour of a sliding support.

DESIGN

Develop a pin-in-slot / collar-on-rod assembly with a sliding element and an attached loading arm.

ANALYSE

Connect degrees of freedom, reaction forces and moments with rigid-body equilibrium.

DEMONSTRATE

Fabricate and observe a physical PLA prototype under progressively applied loads.

My role in the team

The submitted report attributes CAD design and demonstration to me.

MY DOCUMENTED WORK

CAD design contribution

Physical demonstration of the support

Communicating the operating principle

TEAM COLLABORATION

Design briefing and report: MD Zubayer Rahman

Free-body diagram and 3D printing: MD Maruful Hasan Sadik

CAD and demonstration: Bokhtear Uddin Badhon

What does a slider support constrain?

The model represents a collar guided along a horizontal rod.

RESTRICTED MOTION

Vertical translation is restrained.

Rotation is resisted by the model's support arrangement.

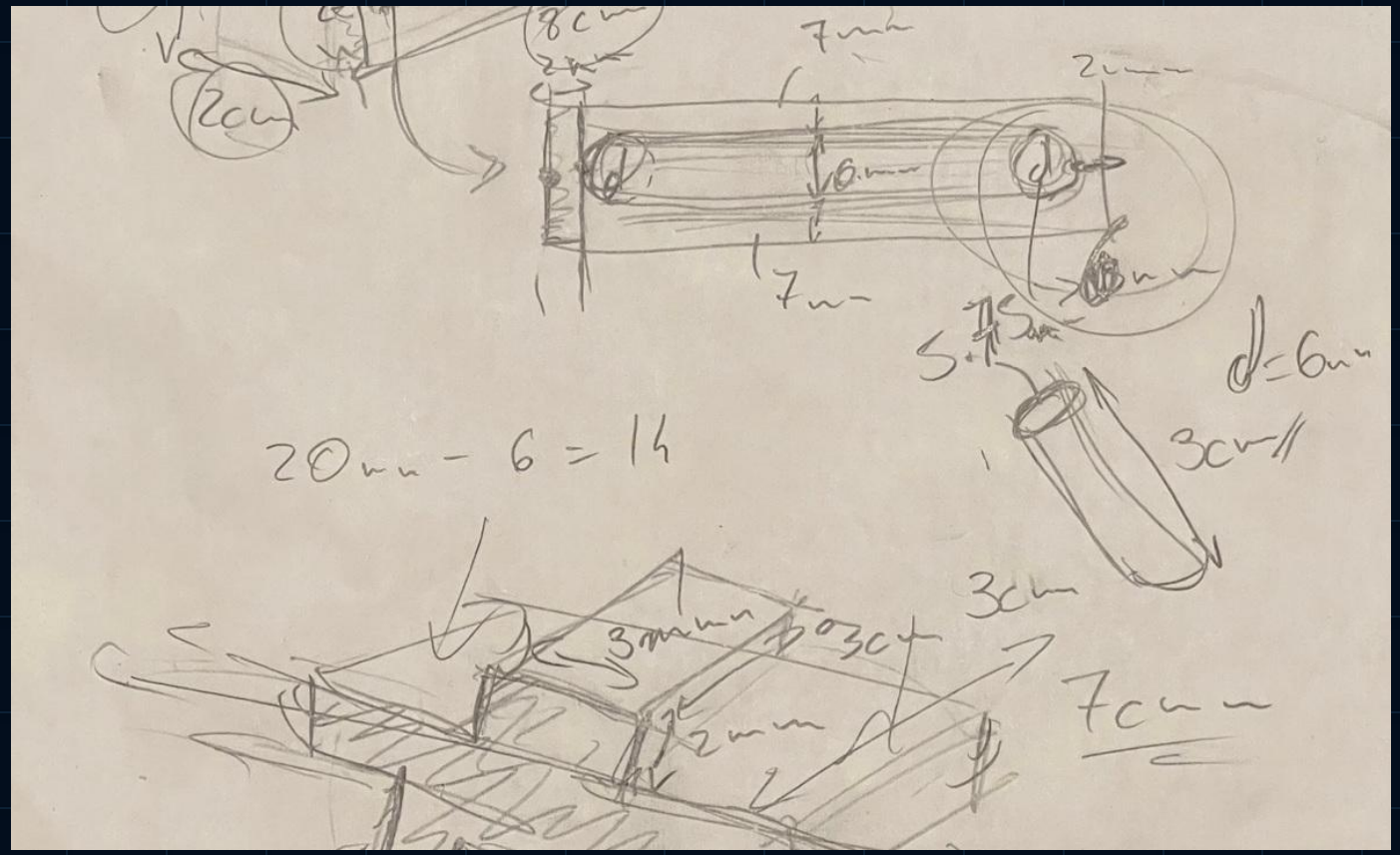
PERMITTED MOTION

Horizontal movement along the rod remains possible.

Ideal analysis assumes negligible friction; the printed model has real friction.

The raw sketch matters

Early drawing records geometry, dimensions and how the assembly could work.



DESIGN THINKING

Explore component geometry

Consider guide length and clearances

Translate a mechanical idea into a manufacturable form

Breaking the assembly into components

The report describes a base, rod, sliding collar, L-shaped beam and weight hanger.

BASE + GUIDE ROD

Approx. 70 mm base length and 6 mm rod diameter reported. The rod defines the translation path.

SLIDER COLLAR

Central bore fits the rod. Clearance and surface finish influence friction and smooth travel.

LOADING ARM

L-shaped beam and weight hanger allow progressively applied vertical loading.

Forces, moments and equilibrium

The goal was to relate the support behaviour to the fundamentals of rigid-body statics.

EQUILIBRIUM

$\Sigma F_x = 0$ • $\Sigma F_y = 0$

$\Sigma M = 0$

For an ideal horizontal slider, horizontal translation remains unrestrained.

PHYSICAL INTERPRETATION

The guide provides constraint reactions to oppose vertical loading and rotation. Actual friction introduces departure from the ideal model.

Designing for physical assembly

The report documents FDM printing with PLA and post-print assembly.

MATERIAL

PLA selected for low cost, printability and sufficient strength for an educational demonstrator.

PRINT SETTINGS

Reported 0.2 mm layer height and 20–30% infill; orientation considered for strength and finish.

ASSEMBLY

Press-fit joints and adhesive where needed; alignment of rod and collar was important.

The prototype — not just a render

A photograph of the physical printed support, retained from the original project material.



WHAT IT SHOWS

- Physical printed components
- Sliding support arrangement
- Visible interfaces and fasteners
- A demonstrator suitable for qualitative observation

What happened under loading?

The submitted report records qualitative observations, not a numerical load-test dataset.

OBSERVED

The collar translated along the rod.

No discernible vertical displacement was reported.

Rotation was resisted.

INTERPRETATION

The observed behaviour was consistent with the intended support constraints.

Minor friction was noticed due to printed surface roughness.

Engineering means recognising constraints

Understanding what a prototype cannot prove is part of a responsible design process.

CURRENT LIMITATIONS

- Non-zero sliding friction
- PLA-limited load capacity
- Printer-resolution and tolerance constraints
- No published quantitative test series

POSSIBLE IMPROVEMENTS

- Polish or improve the guide rod
- Explore bushings or bearings
- Measure sliding force and deflection
- Document repeatable loading conditions

What I learned as a first-year engineer

A small mechanical assembly can reveal major design principles.

ENGINEERING SKILLS

- CAD-based mechanical design
- Constraint and degree-of-freedom reasoning
- Design-for-manufacture awareness
- Prototype demonstration

WORKING PRACTICES

- Collaborating across defined team roles
- Explaining a technical idea clearly
- Learning from real-world friction and tolerances
- Identifying next validation steps

Beyond this project: Engineering for People

A separate first-year design challenge exploring community-centred sustainability in Ladywood, Birmingham.

MY INITIATIVE

Researched the environmental context and proposed a Neighbourhood Solar Sharing System as an individual concept.

TEAM DESIGN PROCESS

Participated in comparing alternative solutions; the team selected the Aerovent heat-recovery ventilation concept. Concept stage only.

Ready to learn. Ready to contribute.

Interested in London-based industrial placements in mechanical design, manufacturing and sustainable engineering.

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